

Personal information

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Research interests

Gravitational waves & binary neutron star mergers Neutron star equation of state
Machine learning & differentiable programming GPU-accelerated Bayesian inference

Professional experience

2023–2027 **PhD Candidate in Physics** Utrecht University
Advisor: Prof. dr. Chris Van Den Broeck
Topic: Bayesian inference on gravitational waves, multi-messenger astrophysics, and the neutron star equation of state

Education

2023 **M. Sc. in Artificial Intelligence** (89.00% summa cum laude) KU Leuven
Advisors: Prof. dr. Tjonnje G. F. Li & Prof. dr. Johan Suykens
2022 **M. Sc. in Theoretical Physics (minor: nuclear physics)** (91.41% summa cum laude with congratulations of the Board of Examiners) KU Leuven
Advisor: Prof. dr. Nikolay Bobev
2020 **B. Sc. in Physics** (74.42% cum laude) KU Leuven
2020 **B. Sc. in Mathematics** (73.92% cum laude) KU Leuven

Publication and citation summary (from inspirehep.net on 29th September, 2026)

	Excluding LVK/ET [†] papers	Including LVK/ET [†] papers
Number of papers:	20	60
Number of citations:	204	3,205
Average citations per paper:	10.2	53.4
h-index:	10	23

[†]LVK = LIGO-Virgo-KAGRA Collaboration, ET = Einstein Telescope Collaboration

Selected publications and preprints ([full list](#), [excl. collaboration papers](#))

F. Gittins, H. Narola, T. Wouters, et al. (2026). Accepted at *Physical Review Letters*.
arXiv: 2606.06376 [gr-qc]
– Prospecting detectability of neutron star resonance modes with the Einstein Telescope

H. Koehn, T. Wouters, et al. (2025). *Astron. Astrophys.* **704**, A55
– Bayesian inference framework (FIESTA) for efficient joint analysis of kilonovae and gamma-ray burst afterglows

T. Wouters et al. (2026). *Phys. Rev. D* **114.4**, p. 043005
– Physics-informed priors constructed with machine learning for gravitational-wave inferences

T. Wouters et al. (2025). *Phys. Rev. D* **112.4**, p. 043037
– Introduces JESTER, an autodifferentiable solver for the Tolman–Oppenheimer–Volkoff equations and GPU-accelerated inference of the neutron star equation of state

T. Wouters et al. (2024). *Phys. Rev. D* **110.8**, p. 083033
– GPU-accelerated inference of gravitational-wave signals from binary neutron star mergers

Collaboration work

- Lead developer of pipeline (JESTER) reviewed within the LIGO-Virgo-KAGRA Extreme Matter working group
- Analyst of the 6th gravitational-wave transient catalog of the LIGO-Virgo-KAGRA collaboration
- Contributed to design study for the Einstein Telescope Collaboration, to be submitted to the board of governmental representatives, using JESTER
- Leading a task force in the Einstein Telescope Nuclear Physics Division on equation-of-state inference systematics, using JESTER

Awards and prizes

2024–2026	SURF small computing grants (3 million CPU core computing hours total)
2023	KU Leuven Short Mobility Grant (EUR 500)
2023	Belgian Physical Society best master thesis in physics (EUR 100)
2022	KU Leuven Physics prize: highest grade among physics master students

Talks

Invited talks and workshops

2027	Differentiable Programming: JAX for Science (<i>upcoming</i>)	Leiden, Netherlands
2027	Constraining the Nuclear Matter Equation of State (<i>upcoming</i>)	Mainz, Germany
2025	Research Seminar	AEI Potsdam, Germany
2025	Accelerated Computing for Application Workflows	SURF, Netherlands
2025	Advanced Computing User Day	SURF, Netherlands
2024	Extreme Matter in Extreme Stars	Leiden, Netherlands
2024	JAXtronomy: JAX for astronomy research	New York, USA

Conference talks

2026	LIGO-Virgo-KAGRA collaboration meeting	Pisa, Italy
2025	European AI for Fundamental Physics Conference	Sardinia, Italy
2025	3rd AIIslands workshop on AI in gravitational waves	Bute, Scotland
2025	XIV Einstein Telescope Symposium	Bologna, Italy
2025	Dutch Physics Society	Lunteren, Netherlands
2024	2nd AIIslands workshop on AI in gravitational waves	Arran, Scotland

Teaching record

Student supervision

Master

2026	Frank Snijder	M. Sc. physics
2025	Marlinde Drent	M. Sc. physics
2025	Oleksii Bevza	M. Sc. data science
2025	Hong Ye Guo	M. Sc. data science

Bachelor

2025	Elin Jakobs	B. Sc. physics
2025	Senna van Os	B. Sc. physics
2024	Frank Snijder	B. Sc. physics
2024	Tom Oeloff	B. Sc. physics
2024	Marlinde Drent	B. Sc. physics

Courses tutored

2024	Master	Machine Learning	Utrecht University
2024	Bachelor	Gravitational waves	Utrecht University
2024	Master	Gravitational waves	Utrecht University
2023	Bachelor	Optics and astronomy	Utrecht University
2020–2022	Bachelor	Introductory mathematics and physics	KU Leuven
2020	Master	Quantum mechanics	KU Leuven

Referee work

- Physical Review D
- Journal of Open Source Software

Public and professional services

2026	Student representative in the Faculty Open Science Team, Utrecht University
2025	Volunteer Einstein Telescope stand at Nerdland, the largest Belgian popular science festival
2024	Outreach lecture on space-time for students of HKU University of Arts, Utrecht
2023	Student representative responsible for the quality of education of the AI master, KU Leuven
